

Satellite quantification of the methane budget and post-fire recovery of a Mediterranean forest fire

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Abstract. The Arischia wildfire (L'Aquila, Italy, 30 July – 13 August 2020) is re-analysed with an open, reproducible Google Earth Engine pipeline built from free optical, thermal and atmospheric satellite data. Intersecting a thematic land-cover map with a 10 m tree-cover map removes 28.3 % of the Copernicus EMS perimeter as non-combustible and leaves 458.0 ha of fuel, of which 309.0 ha burned. A Seiler–Crutzen budget per fuel class gives 14,982 t of biomass consumed and **65.07 t of methane** (16–84 % Monte-Carlo band 42.6–87.6 t), with 24,244 t CO₂, 1,416 t CO and 167 t PM_{2.5}. The carbon mass balance closes at 104.6 %. Against the published analysis of the same fire the methane estimate differs by +9.2 %, obtained without calibration to it, and the optical biomass lies within 11.8 % of that study's independent thermal estimate. MODIS fire radiative energy integrated from daily maxima is an upper bound six to seven times larger than the optical biomass; the ratio implies the fire burned at full intensity for only ~3.5 hours a day. A forward plume model, written down before any Sentinel-5P data were opened, predicts a 0.88 ppb column enhancement — 0.08 σ of TROPOMI's single-sounding precision — so this fire was 25 times below the 2 σ detection floor across all 27 plume geometries tested. Eleven summers of Sentinel-2 resolve exponential canopy recovery with time constants of 8–11 years by severity class. The fire released 47–140 years of the stand's own soil methane uptake; because soil microbial recovery cannot be observed from orbit, the return of that sink is bracketed only as a modelled five-decade window.

1 · WHY METHANE

Carbon dioxide dominates fire emission inventories because it dominates the emitted mass: 93.6 % at Arischia. Methane is 0.25 % of the mass and 0.66 % of the carbon — but one kilogram traps roughly eighty times more heat than a kilogram of CO₂ over twenty years. Weighted over that horizon, the fire's 65 t of methane carries **about 18 %** of its near-term warming. A CO₂-only inventory misses roughly a fifth of the impact.

A second asymmetry makes methane more than a footnote. Temperate forest soils are the main biological land sink for atmospheric methane: methanotrophic bacteria in the organic horizon oxidise 1.5–4.5 kg CH₄ ha⁻¹ yr⁻¹ (Dutaur & Verchot 2007). A fire therefore acts twice — it emits a pulse and it removes the mechanism that would have taken methane back out of the air. Over 309 ha, the lost uptake is 0.46–1.39 t CH₄ yr⁻¹, so the two-week pulse equals **47–140 years** of this forest's own removal capacity: a *methane debt*.

This note asks three questions. (i) How much methane was released, with what honest uncertainty? (ii) Could a satellite have seen it? (iii) When does the forest — and its methane sink — recover?

2 · STUDY AREA AND REFERENCE

The fire burned in the central Apennines above Arischia (42.4° N, 13.3° E), on slopes averaging 23°, in a stand dominated by coniferous forest (60 % of the fuel area). Copernicus EMS activation **EMSR449** provides an operational burn delineation of 638.99 ha, used here only as reference geometry.

Laneve & Pampanoni (2021) analysed the same event with Sentinel-2, PlanetScope and thermal fire-radiative-power data. Their Sentinel-2 budget (59.60 t CH₄) and their polar-orbit thermal budget (16,751 t biomass) serve as **independent anchors**: they are compared against at the end, never used to tune any parameter.

3 • DATA

Domain	Sensor / product	Resolution	Role
Optical	Sentinel-2 MSI L2A	10–20 m, 5-day	burn severity, moisture, eleven-year recovery
Thermal	MODIS Terra + Aqua, MOD14A1/MYD14A1	1 km, ~4 looks/day	fire radiative power and energy
Atmospheric chemistry	Sentinel-5P TROPOMI CH ₄ and CO	~5.5 × 7 km, daily	detectability test
Land cover	CORINE 2018, ESA WorldCover 2020	100 m, 10 m	fuel type ∩ fuel extent
Biomass	ESA CCI AGB v6 (2019)	100 m	fuel load per class
Weather	ERA5-Land	~9 km	pre-fire temperature and rainfall
Reference	Copernicus EMS EMSR449	vector	perimeter

All processing runs server-side in Google Earth Engine from a single Colab notebook; every number is written to one results register at the moment it is computed.

4 • METHODS

Full formulas and constants are in [03_METHODS.md](#); the essentials:

- **Fuel.** CORINE forest classes (311, 312, 313, 324) intersected with WorldCover tree and shrub cover. No tuned parameter.
- **Severity.** $NBR = (B8 - B12) / (B8 + B12)$, $dNBR = NBR_{pre} - NBR_{post}$; burned where $dNBR \geq 0.16$; four classes at 0.27, 0.44, 0.66 (Key & Benson 2006).
- **Budget.** $BB = \sum A \cdot AGB \cdot BE$ per fuel class; $E_x = \sum BB \cdot EF_x$ with the class-specific emission factors of Laneve & Pampanoni (2021). Uncertainty by 200,000 Monte-Carlo draws (area $\pm 5\%$, AGB $\pm$ class σ , BE $\pm 30\%$, EF $\pm 40\%$). A severity-dependent combustion completeness ($\beta = 0.20 / 0.35 / 0.50 / 0.65$) is run as a sensitivity test.
- **Thermal energy.** $FRE = \int FRP \, dt$ from daily MODIS maxima, $BB = 0.368 \cdot FRE$ (Wooster et al. 2005). Because daily maxima are treated as sustained, this is an upper bound; its ratio to the optical biomass is interpreted as a **diurnal duty cycle**.
- **Detectability.** Integrated-mass-enhancement balance run forwards: $\tau = L/U$, $IME = Q \cdot \tau$, $\Delta XCH_4 = (IME/M) \cdot N_A / (A \cdot N_{air})$. The prediction was saved with a timestamp before TROPOMI data were queried, then swept over 27 plume geometries.
- **Recovery.** One peak-season NBR composite per summer, 2016–2026, per severity class and for an unburnt control forest; $\ln(NBR_{base} - NBR)$ fitted linearly in time (exponential recovery). Full recovery is declared when the deficit falls below the control forest's interannual σ ; the methane-sink window applies an assumed soil-microbial lag of 1.3–2.0× to the canopy timeline.

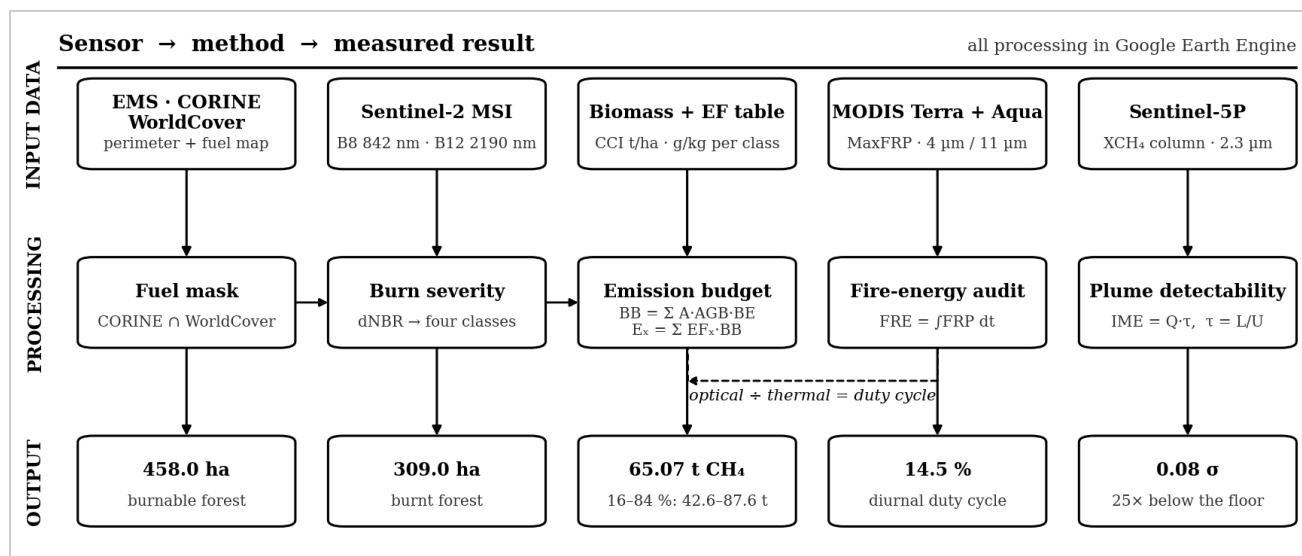


Figure 1. Sensor → method → measured result. All processing in Google Earth Engine.

5 · RESULTS

5.1 Areas and severity

Quantity	ha	% of perimeter
EMS perimeter	638.99	100.0
removed as non-fuel	181.0	28.3
burnable forest	458.0	71.7
burned forest (dNBR ≥ 0.16)	309.0	48.4
— Low / Moderate / High / Very high		95.7 / 79.1 / 52.3 / 81.9

5.2 Emission budget

Species	Mass [t]	t ha ⁻¹	% of mass	Implied EF [g kg ⁻¹]
CO ₂	24,244	78.5	93.6	1,618
CO	1,416	4.58	5.47	94.5
PM _{2.5}	167	0.54	0.65	11.2
CH₄	65.07 (42.6–87.6)	0.211	0.25	4.34
Biomass consumed	14,982	48.5	—	—

The asymmetric band is the signature of a product of uncertain terms. The emission factor dominates the variance; burned area contributes least — so better imagery would not narrow the result, but a field-measured emission factor would. Distributing combustion completeness by severity instead of by fuel class changes methane by only –3.0 % (63.12 t).

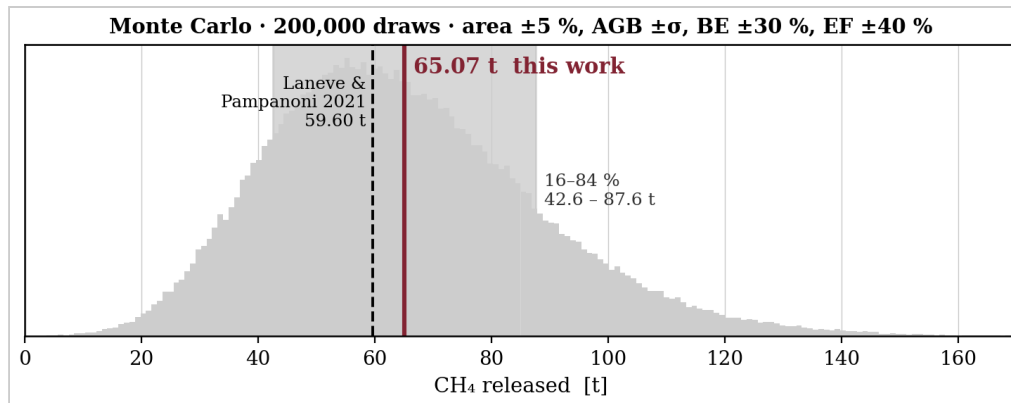


Figure 2. Monte-Carlo distribution of methane released, with the 16–84 % band and the reference value.

5.3 Consistency checks

Check	This work	Reference / closure	Agreement
Methane vs Laneve & Pampanoni (2021), Sentinel-2 route	65.07 t	59.60 t	+9.2 %, inside the 16–84 % band
Optical biomass vs their thermal (FRP, polar-orbit) route	14,982 t	16,751 t	11.8 %
Carbon emitted vs carbon available (BB × 0.47)	7,368 tC	7,042 tC	104.6 %

The comparison also exposes an internal inconsistency in the reference: its Sentinel-2 budget implies an emission factor for CO₂ of 1,390 g kg⁻¹ (23,329 t ÷ 16,783 t), below every class value in its own emission-factor table (1,592–1,639 g kg⁻¹). The present budget implies 1,618 g kg⁻¹, inside that range — a ratio of 1.164 that follows directly from the table.

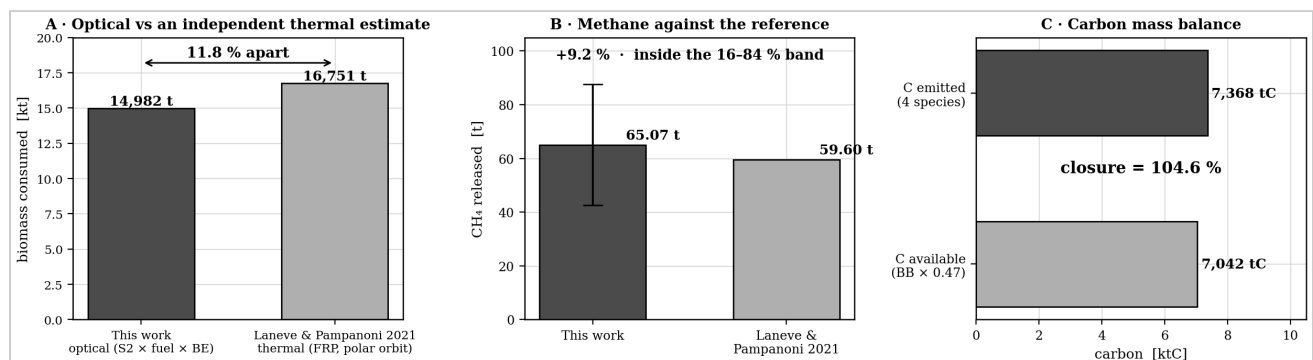


Figure 3. (A) optical biomass against the reference study's thermal estimate; (B) methane against the reference; (C) carbon mass balance.

5.4 How long the fire actually burned

MODIS saw the fire on five days, peaking at 4,115 MW. Integrating the daily maxima as if sustained gives 279.8 TJ and an upper-bound biomass of 102,976 t — 6.9 times the optical estimate. Read the other way round, the optical budget is 14.5 % of that bound: the fire released its energy as if it had burned at peak intensity for about **3.5 hours a day**, consistent with a strongly diurnal Mediterranean fire.

5.5 Was the methane visible from orbit?

The acute-phase emission rate is 6.5 t CH₄ day⁻¹ (60 % of the methane in the first six days). At the nominal geometry (L = 20 km, U = 3 m s⁻¹, A = 100 km²) this gives $\Delta X_{CH_4} = 0.88$ ppb — **0.08 σ** of TROPOMI's ~11 ppb single-sounding precision. Detection at 2σ would need 165 t day⁻¹; across the 27-geometry sweep the floor ranges from 27 to 1,098 t day⁻¹ and the best achievable significance is 0.47 σ. The fire was **25 times below the floor**. In the event, TROPOMI returned no valid methane retrieval over the site during the fire window (33

valid days for CO), so the conclusion rests on the forward model — which is exactly why the model was written down first.

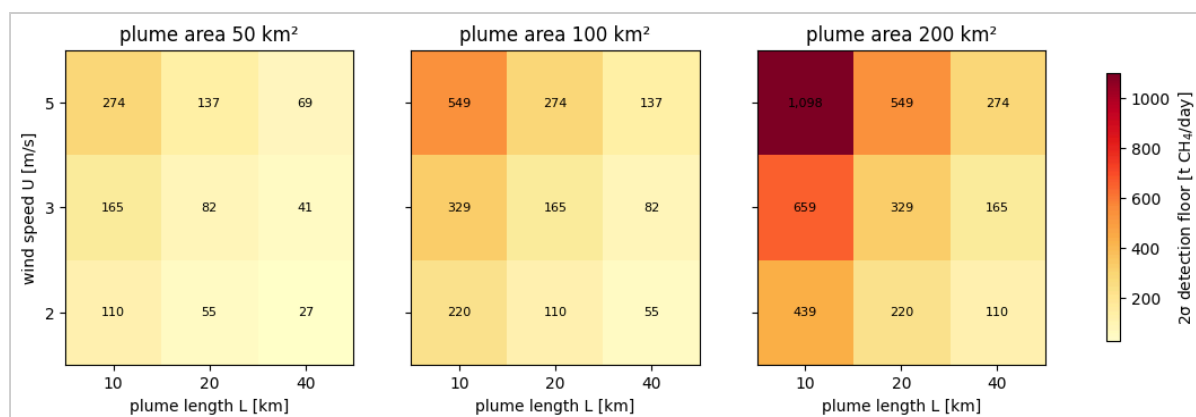


Figure 5. 2σ TROPOMI detection floor [t CH₄ day⁻¹] across 27 plume geometries. The fire emitted 6.5 t day⁻¹.

5.6 Recovery

Severity	Recovered by 2026	k [yr]	90 % canopy	Full canopy	CH ₄ sink (modelled)
Low	66.7 %	10.9	2039	2057	2045 – 2094
Moderate	59.4 %	8.9	2038	2053	2043 – 2085
High	51.9 %	8.3	2038	2052	2044 – 2083
Very high	35.8 %	8.3	2041	2055	2047 – 2089

The unburnt control drifted by −1.4 % over the decade with an interannual σ of 0.0108 NBR, so the recovery signal is not climate. Severity sets how deep the hole is, not how fast the forest climbs out: the time constants are similar, the starting depths are not. The pre-fire moisture test (NDMI −0.66 σ, 180.6 mm of rain in June–July) indicates the stand was not anomalously dry — an ignition-limited rather than a drought-primed fire.

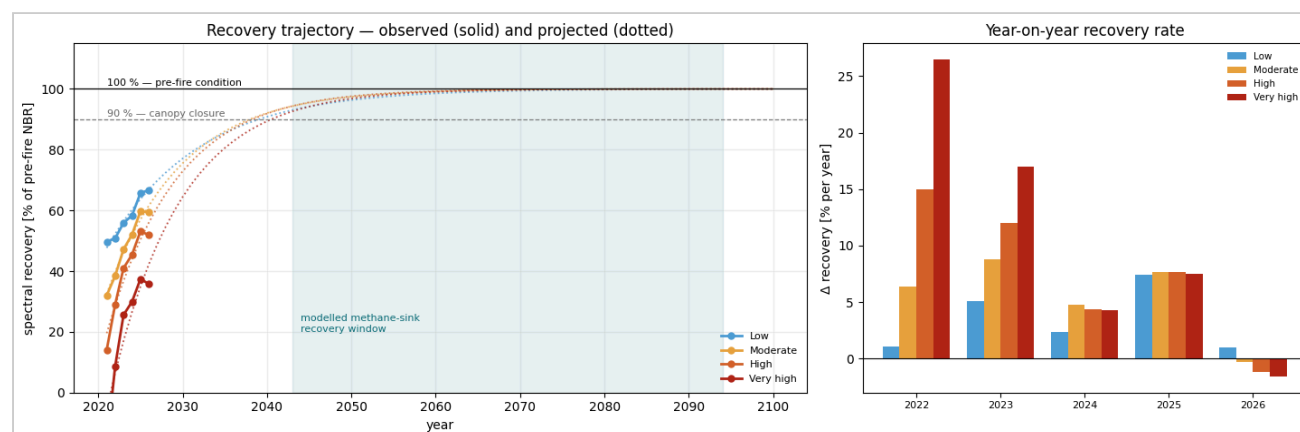


Figure 4. Spectral recovery by severity class — observed (solid) and projected (dotted), with the modelled methane-sink window.

6 · LIMITS

1. Emission factors, not imagery, limit the methane estimate (± 40 %).
2. The MODIS energy integral is an upper bound here and is not used as a second biomass estimate.
3. NBR is a spectral proxy for canopy, not a measure of structure or carbon.
4. The methane-sink window is modelled; methanotroph recovery is invisible to every satellite band, and closing that range needs soil flux chambers or eddy-covariance measurements along a severity gradient.

7 • CONCLUSIONS

1. **Quantification** — 65.07 t CH₄ (42.6–87.6 t) from 309 ha, within 9.2 % of an independent study and with a closing carbon balance.
2. **Detectability** — a fire of this size sits 25× below what a daily global methane satellite can see, predicted before looking.
3. **Recovery** — the canopy returns within two to four decades; the methane sink is bracketed only as a modelled 2043–2094 window, and 47–140 years of uptake are owed back.

8 • NEXT STEPS

- **Structure, not only colour** — GEDI lidar canopy height matched to severity classes, and Sentinel-1 SLC interferometric coherence processed in ESA SNAP.
- **Data fusion** — optical, radar and lidar in one recovery model; VIIRS 375 m fire radiative power for a sharper thermal route.
- **Prediction** — Gaussian-process regression in log-deficit space for recovery with credible intervals.
- **Sensors that could see it** — EnMAP, PRISMA and GHGSat now; Copernicus CO2M next.
- **More fires** — the same notebook run on larger Mediterranean events to map where the detection floor is crossed.

ACKNOWLEDGEMENTS

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